

Juan Antonio Monleón de la Lluvia

Nuclear Engineer · PhD in Nuclear Physics

CONTACT

juanmonleon96@gmail.com
[GitHub - juanmonleon](#)
[ORCID 0009-0007-9123-1806](#)
[LinkedIn - Juan A. Monleón](#)
[+34 634 683 078](#)

EDUCATION

PhD in Nuclear Physics
Université Paris-Saclay · ED
PHENIICS 2023-2026

**Double Master in Industrial Engineering
and Nuclear Science and Technology**
Polytechnic University of
Madrid 2021-2023

**Master in Management and Direction of
SMEs**
EFEM Business School 2024-2025

**Bachelor in Industrial Technology
Engineering**
Polytechnic University of
Madrid 2017-2021

TECHNICAL SKILLS

Nuclear Codes: MCNP, Serpent, OpenMC,
NJOY, ADVANTG
Programming: Python, Git, Claude Code
Documentation: LaTeX, Microsoft 365

ABOUT ME

Nuclear engineer with a PhD in Nuclear Physics from Université Paris-Saclay, working on nuclear data and on the sensitivity and uncertainty analysis of neutronic simulations, from the evaluation of covariances to their propagation in shielding and reactor-dosimetry calculations. Proactive and quick to adapt to new codes, tools and problem domains.

RESEARCH EXPERIENCE

Doctoral Research in Nuclear Data & Neutronics

ASNR, France · Université Paris-Saclay Nov 2023 - Oct 2026

Thesis: “Nuclear-Data Uncertainty Quantification for PWR Vessel Aging with a ⁵⁶Fe Elastic-Scattering Evaluation” (defended 1 October 2026)

- Sensitivity analysis and Monte Carlo sampling to propagate nuclear-data uncertainties; re-evaluated ⁵⁶Fe elastic angular distributions and covariances from EXFOR data
- Applied variance-reduction techniques (ADVANTG weight windows) to improve efficiency in deep-penetration shielding calculations
- Supervised a Master's student (Mar–Sep 2025) on nuclear-data uncertainty quantification in PWR source-term calculations

Master's Thesis Research (MII)

UPM, Spain Jun 2023 - Oct 2023

- Developed “Enhancing the EXFOR nuclear data library with Machine Learning Techniques”
- Implemented outlier detection methods for proton-induced reactions

Research Internship — Master's Thesis (MUCTN)

SCK CEN, Belgium Sep 2022 - Jun 2023

- Completed Master's thesis titled “Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques”
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Internship

ENUSA Advanced Industries, Spain Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

PUBLICATIONS

- J. A. Monleon de la Lluvia, M. Brovchenko, D. Rochman, E. Dumonteil, “Towards Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations”, *Nuclear Science and Engineering*, 2025. DOI: 10.1080/00295639.2025.2510048 · [pre-print](#) 2025

LANGUAGES

Spanish Native language

English C2 Level

French C1 Level

AWARDS

Best Academic Report for Nuclear Master
Polytechnic University of Madrid (2022-2023)

Best Master Thesis Award
SCK CEN, Belgium (2023)

Best Poster Award
Journées des Thèses of IRSN (2024) - [View Poster](#)

COMMUNITY

JEFF Community

Active member · Joint Evaluated Fission and Fusion File (OECD/NEA)

INDEN Community

Member · International Nuclear Data Evaluation Network (IAEA)

SINBAD Community

Member · Shielding Integral Benchmark Archive & Database (OECD/NEA)

REFERENCES

Dr. Mariya Brovchenko

ASNR, France
mariya.brovchenko@asn.fr

Dr. Dimitri Rochman

Nagra, Switzerland
dimitri.rochman@nagra.ch

Prof. Óscar Cabellos

Universidad Politécnica de Madrid, Spain
oscar.cabellos@upm.es

CONFERENCES & PRESENTATIONS

FIRST AUTHOR

- “Evaluation of elastic angular distribution covariances for ^{56}Fe from EXFOR experimental data” — WONDER 2026, Aix-en-Provence Jun 2026
- “Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to ^{56}Fe ” — PHYSOR 2026 Apr 2026
- “Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment” — ND 2025, Madrid Jun 2025
- “Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems” — ANS Winter Conference RPSD, Orlando Nov 2024

CO-AUTHORED

- “Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6” — PHYSOR 2026 Apr 2026
- “Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations” — PHYSOR 2026 Apr 2026

SOFTWARE & TOOLS

KIKA — Desktop Workspace for Nuclear Data

Author & lead developer 2025 - Present

- Opens ENDF, ACE, EXFOR and SDF files, drives a local NJOY installation, reads MCNP and Serpent runs, and carries the results into nuclear-data sampling, sensitivity and uncertainty analysis, all in a single desktop application
- Also author of [kika-nd](#), the open-source Python library (PyPI) I wrote to provide the computational core behind the application

TRAINING

- OpenMC Course — NEA, Paris Dec 2025
- NJOY Course — NEA, Paris Dec 2024
- Sensitivity Analysis Summer School (SAMO) — UNIPR, Parma Jun 2024
- Serpent Bootcamp — NEA, Paris Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" — GREAT PIONEER Dec 2022